

Alkylphenols in human milk and their relations to dietary habits in central Taiwan.



The aims of this study were to determine the concentrations of 4-nonylphenol (NP) and 4-octylphenol (OP) in 59 human milk samples and to examine related factors including mothers' demographics and dietary habits. Women who consumed over the median amount of cooking oil had significantly higher OP concentrations (0.98 ng/g) than those who consumed less (0.39 ng/g) ($P < 0.05$). OP concentration was significantly associated with the consumption of cooking oil (beta = 0.62, $P < 0.01$) and fish oil capsules (beta = 0.39, $P < 0.01$) after adjustment for age and body mass index (BMI). NP concentration was also significantly associated with the consumption of fish oil capsules (beta = 0.38, $P < 0.01$) and processed fish products (beta = 0.59, $P < 0.01$). The food pattern of cooking oil and processed meat products from factor analysis was strongly associated with OP concentration in human milk ($P < 0.05$). These determinations should aid in suggesting foods for consumption by nursing mothers in order to protect their infants from NP/OP exposure. 2010 Elsevier Ltd. All rights reserved.